

Python + ML + AI Mastery Guide

A structured guide to master Python, Machine Learning, Deep Learning, NLP, and AI Engineering concepts with direct learning links.

Python Fundamentals

- **Python Basics**

<https://www.youtube.com/watch?v=rfscVS0vtbw>

- **Data Types**

<https://docs.python.org/3/tutorial/introduction.html>

- **Functions**

<https://realpython.com/defining-your-own-python-function/>

- **OOP Basics**

https://www.youtube.com/watch?v=Ej_02ICOlgs

- **List Comprehensions**

<https://realpython.com/list-comprehension-python/>

- **Virtual Environments**

<https://docs.python.org/3/library/venv.html>

NumPy + Pandas

- **NumPy**

<https://numpy.org/learn/>

- **Broadcasting**

<https://numpy.org/doc/stable/user/basics.broadcasting.html>

- **Vectorization**

<https://realpython.com/numpy-array-programming/>

- **Pandas**

https://pandas.pydata.org/docs/getting_started/index.html

- **Data Cleaning**

<https://www.kaggle.com/learn/data-cleaning>

- **GroupBy**

<https://pandas.pydata.org/docs/reference/groupby.html>

- **Joins**

<https://realpython.com/pandas-merge-join-and-concat/>

Data Visualization

- **Matplotlib**

<https://matplotlib.org/stable/tutorials/index.html>

- **Seaborn**

<https://seaborn.pydata.org/tutorial.html>

- **Plotly**

<https://plotly.com/python/>

Statistics + Math

- **Statistics Basics**

<https://www.khanacademy.org/math/statistics-probability>

- **Probability**

<https://www.youtube.com/watch?v=uzkc-qNV0Ok>

- **Linear Algebra**

<https://www.3blue1brown.com/topics/linear-algebra>

- **Vectors & Matrices**

<https://www.khanacademy.org/math/linear-algebra>

- **Eigenvalues**

<https://www.youtube.com/watch?v=PFDu9oVAE-g>

Machine Learning

- **Scikit-learn**

<https://scikit-learn.org/stable/>

- **Train/Test Split**

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

- **Linear Regression**

<https://developers.google.com/machine-learning/crash-course/linear-regression>

- **Decision Trees**

<https://scikit-learn.org/stable/modules/tree.html>

- **Random Forests**

<https://www.ibm.com/topics/random-forest>

- **Gradient Boosting**

<https://xgboost.readthedocs.io/en/stable/>

- **K-Means**

<https://scikit-learn.org/stable/modules/clustering.html#k-means>

- **PCA**

<https://scikit-learn.org/stable/modules/decomposition.html#pca>

Deep Learning

- **Neural Networks**

<https://www.youtube.com/watch?v=aircAruvnKk>

- **Activation Functions**

<https://www.geeksforgeeks.org/activation-functions-neural-networks/>

- **Loss Functions**

<https://towardsdatascience.com/common-loss-functions-in-machine-learning-46af0ffc4d23>

- **Backpropagation**

<https://www.youtube.com/watch?v=llg3gGewQ5U>

- **PyTorch**

<https://pytorch.org/tutorials/>

- **TensorFlow**

<https://www.tensorflow.org/tutorials>

NLP + LLMs

- **NLP Basics**

<https://www.coursera.org/learn/language-processing>

- **Tokenization**

https://huggingface.co/docs/transformers/tokenizer_summary

- **Embeddings**

<https://www.pinecone.io/learn/vector-embeddings/>

- **Transformers**

<https://jalammar.github.io/illustrated-transformer/>

Computer Vision

- **Computer Vision Basics**

<https://opencv.org/courses/>

- **CNN Intuition**

https://www.youtube.com/watch?v=YRhxdVk_sls

MLOps + Production

- **Model Versioning**

<https://dvc.org/>

- **Experiment Tracking**

<https://mlflow.org/>

- **Model Serving**

<https://fastapi.tiangolo.com/>

- **MLOps Basics**

<https://madewithml.com/courses/mlops/>

AI Ethics

- **Bias in AI**

<https://developers.google.com/machine-learning/fairness-overview>

- **Data Leakage**

<https://machinelearningmastery.com/data-leakage-machine-learning/>

- **Model Explainability**

<https://christophm.github.io/interpretable-ml-book/>

Final Advice

Focus on projects, consistency, and practical implementation. The fastest way to learn AI is by building.